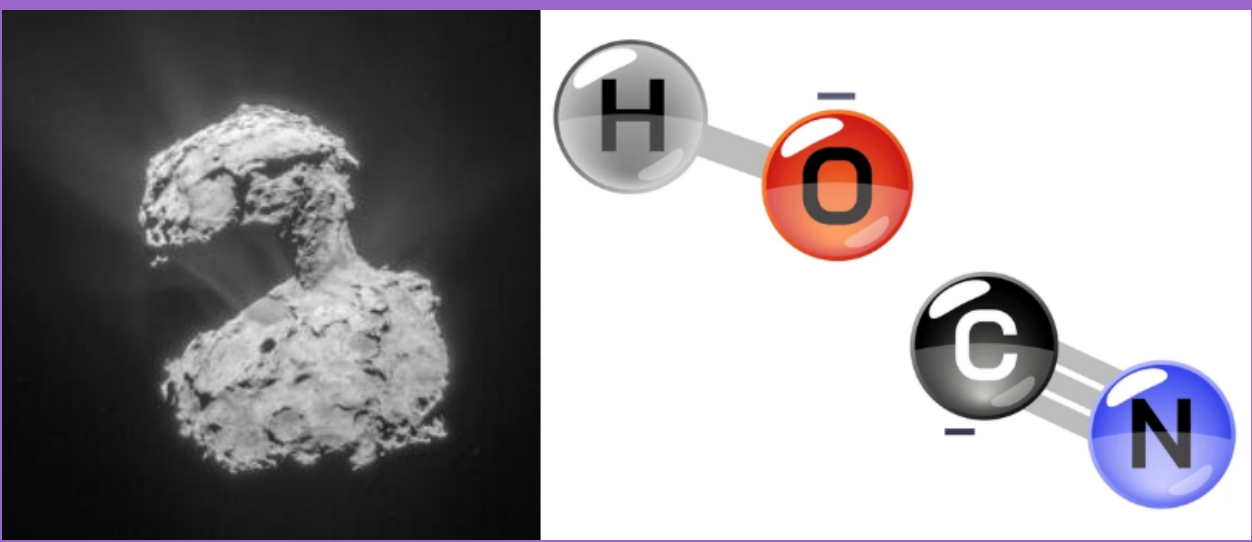




Production Rates of Cyanogen in Comet 67P's Coma

Ava Lee Allen¹ Advisors: Dr. Lori Feaga¹, Dr. Tony Farnham¹

¹Department of Astronomy, University of Maryland, College Park



Abstract

Through observations of **Comet 67P/Churyumov-Gerasimenko** (CG) acquired using the Lowell Discovery Telescope, the production of cyanogen (CN) with respect to the production of hydroxide (OH) was determined at various heliocentric distances throughout the comet's 2021-22 apparition. Absolute fluxes were used as inputs to the Haser Model to derive production rates for CN and OH and calculate logarithmic production rate ratios of CN/OH.

Introduction

- The **coma** or atmosphere around the center nucleus of a comet is made of material that has sublimated off of the surface as it approaches the Sun and **perihelion** is the point in the comet's orbit where it is closest to the Sun and is generally most productive.
- The molecules in the coma that sublimate directly off of the comet's nucleus are known as parent molecules and flow radially outward until they are struck by a solar photon and split into new daughter molecules, a process known as **photodissociation**.
- Often the **daughter molecules** are more readily observed from ground-based telescopes and the composition of the comet nucleus must be inferred from them.
- The **Haser Model** is a mathematical model that calculates total production rates in cometary comae based on their absolute flux and known properties of the molecule in question.
- In previous comet studies, H₂O, OH, HCN, and CN are well established constituents in the coma, with OH and CN the expected daughter products of H₂O and CN, respectively. However, there is **evidence in some comets that HCN is not the sole parent of CN**.
- Comet 67P/Churyumov-Gerasimenko** is a Jupiter-family comet that was discovered in 1969. The *Rosetta* mission studied comet CG in detail from 2014-2016. The comet is well characterized and has had many precursor and subsequent ground-based observational campaigns. We use CG for our study of the parent to daughter relationship of HCN:CN.

Data

- The observations were made using the 4.3 meter **Lowell Discovery Telescope** (LDT) in Arizona, specifically with the **Large Monolithic Imager** (LMI), the telescope's optical camera that is equipped with **HB Narrowband Filters** to specifically target the OH and CN gas.
- Standard stars** are stars with well known magnitudes that we can use to determine how the how the atmosphere has affected our observations and calibrate the data accordingly.

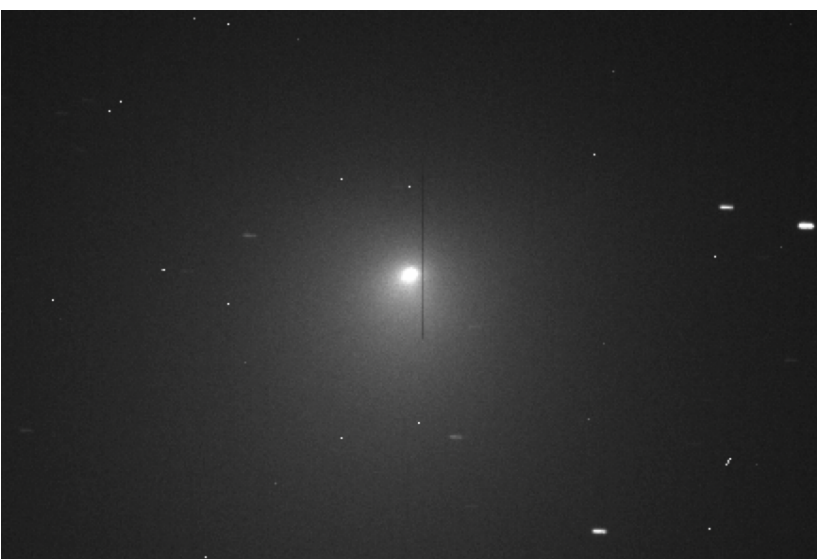


Figure 1. CG through CN filter on 20211109

Determining Counts and Instrumental Magnitude

The number of counts (electrons detected per pixel during an image exposure) can be determined over a radius around the comet through **aperture photometry**. This number can then be converted into an instrumental magnitude for all observations of the comet and each standard star in OH and CN using this formula where C is the counts within the target aperture and T_e is the exposure time of the image:

$$M_i = 25 - 2.5 \log(C/T_e) \quad (1)$$

Converting to Absolute Magnitude

To determine the absolute magnitude of the comet through the CN filter, we must correct the instrumental value for the effects of **atmospheric extinction**. These are determined through a fitting process on the standard stars using these formulas:

CN:

$$M_a = M_i + \Delta m + (E \cdot A) \quad (2)$$

where M_i is the instrumental magnitude and A is the airmass of the object during the observation. The fitted parameters are Δm and E which physically parameterize the extinction and correspond to the intercept and slope of the linear fit.

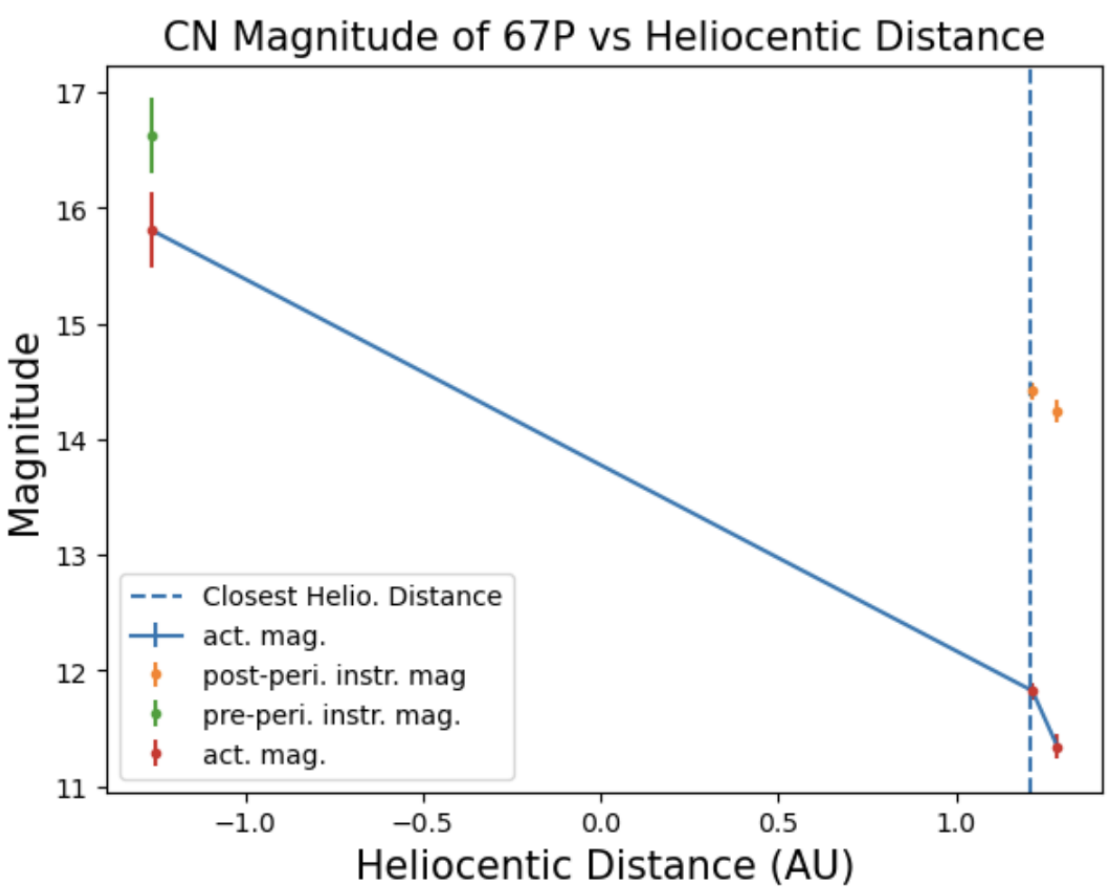


Figure 2. Magnitude of CG through the CN Filter

OH:

$$M_0 = M_i - G_{ROH} - E_{AOH}X - G_{OOH} \quad (3)$$

where G_{ROH} is the correction calculated for the **Rayleigh** component, G_{OOH} is the correction calculated for the **Ozone** component, and E_{AOH} is the correction calculated for the **Aerosol** component.

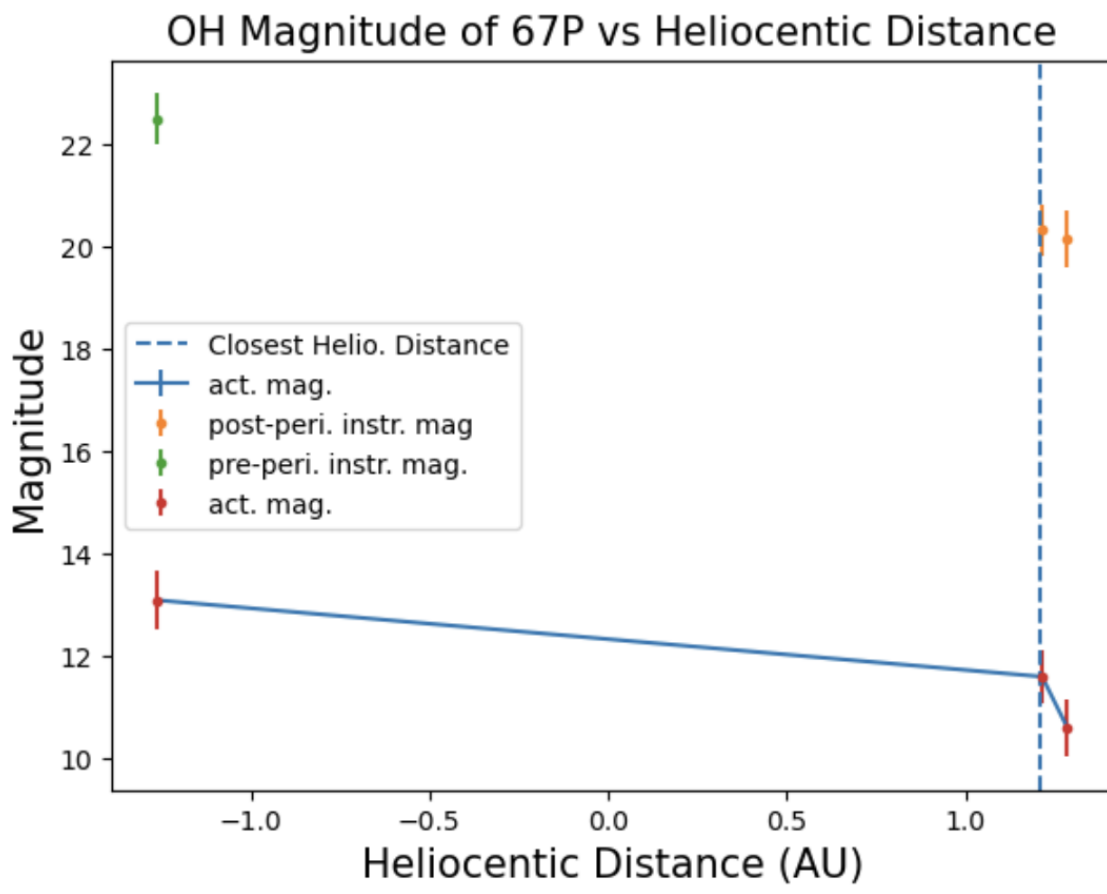


Figure 3. Magnitude of CG through the OH Filter

Determining Absolute Flux

Absolute magnitudes were converted into absolute fluxes using the formula below:

$$F_{abs} = F_0 \cdot 10^{-0.4M_a} \quad (4)$$

where F_0 is $10.56 \cdot 10^{-9}$ for OH and $8.6 \cdot 10^{-9}$ for CN and M_a is the absolute magnitude of the comet. These fluxes were corrected for the portion of the gas band emission that was not captured by the bandpass of the filter. This was done by dividing F_{abs} by a factor of γ , which is $1.812 \cdot 10^{-2}$ for CN and $1.698 \cdot 10^{-2}$ for OH.

Production Rates using the Haser Model

With the absolute flux of CG determined, the Haser Model was utilized to estimate the comet's production rates. This model depends on absolute flux, CN and OH scale lengths, g-factors for OH and CN, as well the radial distance from the nucleus of the comet (determined using the pixel scale of the instrument and the distance from the observer to the comet). The Haser code solves for the **production rate** (Q) in the equation below:

$$n_\rho = (Q/4\pi\nu\rho^2)(l_d/l_p - l_d)(e^{-\rho/l_p} - e^{-\rho/l_d}) \quad (5)$$

where ρ is the projected distance from the comet's nucleus, l_d and l_p are, respectively, the daughter and parent scale lengths, ν is the expansion velocity, and n_ρ is the column density calculated within the program.

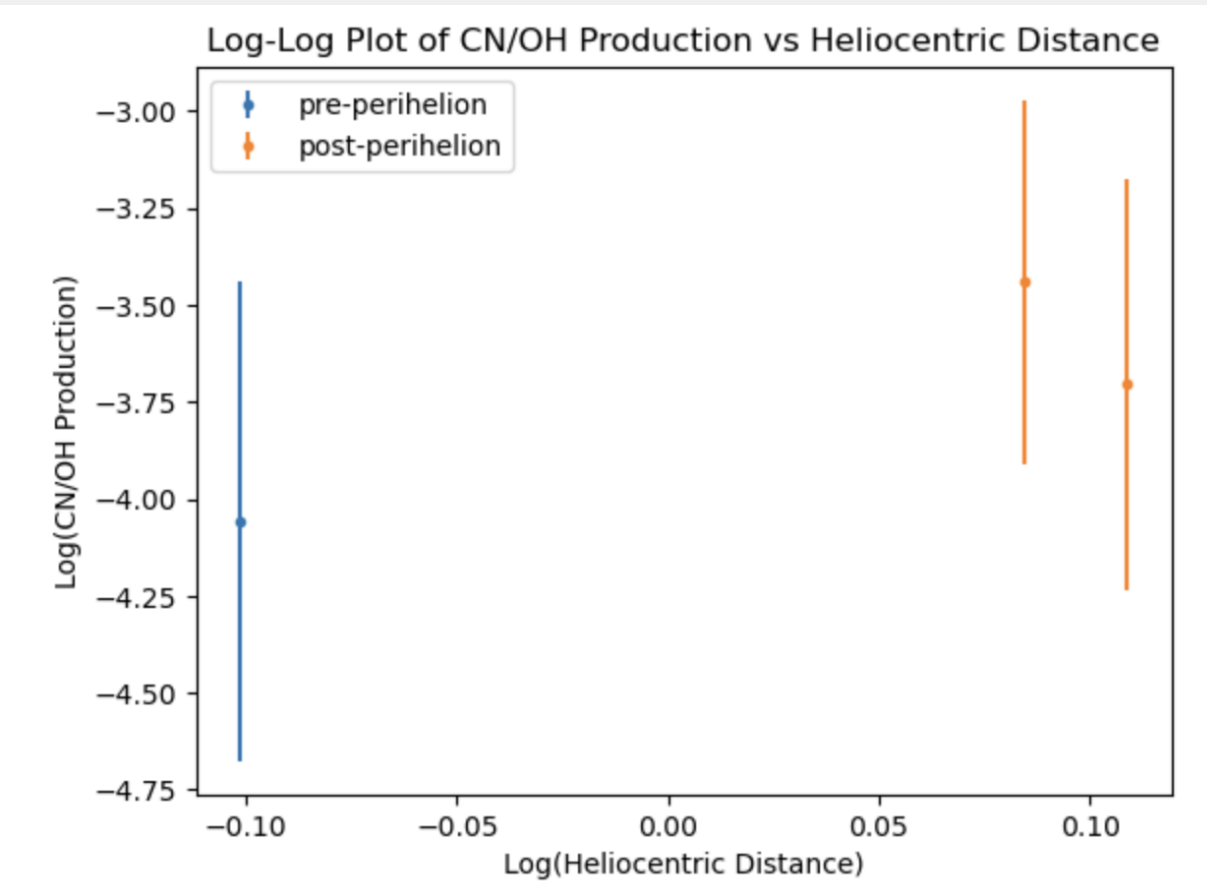


Figure 4. CN/OH Production Rate Ratios

The ratios shown in Figure 4 indicate how much CN is produced with respect to OH, which is dependent on heliocentric distance. The production rate ratios shown in this plot are -4.06 ± 0.62 for pre-perihelion, and -3.44 ± 0.47 for immediately post-perihelion.

Conclusions

These results compared to those found in other studies **do not conclusively imply that there must be another parent of CN** within the coma of comet 67P/C-G.

Further studies must be done to account for rotational and seasonal variability.

References



Full Paper and Sources